External Mentor Project

Agricultural Harvesting Robot

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1. Market research analysis

1.1 Market status and trend of agricultural robots

Agricultural robot is the application of robots in agricultural production, is a kind of unmanned automatic operation machinery that can be controlled by different program software to adapt to various operations, and can feel and adapt to crop types or environmental changes. With the development of science and technology and the continuous promotion of the government, the application of agricultural robots in agricultural production in China is increasingly widespread. As early as 2019, China issued policy documents such as the Action Plan for Digital Rural Development (2022-2025), which made important deployment of national digital agricultural projects. According to statistics, in 2019, China's agricultural robot production increased from 32.5 million units in 2015 to 17,000 units, and the demand increased from 35.15 million units in 2015 to 17,273 units. In the 2023 government work report, innovation and promotion of agricultural robots should be accelerated on the specific landing of agricultural robots, and the output of agricultural robots in China is expected to be about 31,193 units this year, and the demand is about 31,473 units. Driven by the policy, investment and financing in the field of smart agriculture in China from 2015 to 2022 is more active, and there are 20-30 companies invested every year, including Tencent, IDG Capital, Hillhouse Venture Capital, Shunwei Capital and other head institutions.

1.2 Application scenario and function analysis of agricultural robots

Just make sure you have the correct citation format.Robots are widely used in the agricultural field, including breeding and raising seedlings, sowing, irrigation, picking and harvesting, sorting, inspection and so on. For example, through 5G, image recognition technology and big data systems, the picking robot can instantly calculate the distance between the crop fruit and the robot human, and convey the relevant instructions to the robot arm to achieve rapid picking; Weeding robots can accurately identify weeds and crops through autonomous analysis or human-computer interaction, and then wave knives to accurately remove weeds. Taking a picking robot as an example, typical functions required for a picking operation include: vision systems, motion planning (navigation methods), and grasping strategies. The robot first scans the orchard through the vision system, confirms and locates the target (ripe fruit), and then plans the movement route through navigation, so that the mobile platform equipped with the robot manipulator approaches the target, and finally grabs and cuts.

1.3 Market opportunities and challenges of agricultural robots

At the same time of scientific and technological progress, the requirements for the degree of automation of agricultural robots have also increased. At present, autonomous driving technology has become the focus of the development of agricultural robots, and its popularity will greatly promote the improvement of agricultural production efficiency, and the market size of agricultural robots will be further expanded. The development of agricultural robot in China has achieved good results, but when the robot enters the new application scenario, it still faces many new challenges. Technical challenges include the lack of accuracy in visual recognition and spatial positioning of complex environments, and the low maturity of machine control decision systems. At present, autonomous operating robots are mostly applied to fixed scenes, and in the case of unstructured environment, the environment itself has great uncertainty, and the change of each position of the robot is a process of re-establishing coordinates, which requires new correction, measurement, sensing and control. Therefore, robot development is the biggest challenge. Secondly, the challenges facing the business are: the overall utilization rate is low, and the price is expensive, which makes it difficult for ordinary farmers and operators to bear. Most agricultural robots can only operate for a certain link in the agricultural production process, and are accompanied by strong seasonal characteristics, resulting in an indirect increase in production costs.

1.4 The result of the market research analysis

Under the background of the continuous development of automation, digitalization and robots, the application of agricultural robots is an inevitable trend in the development of modern agriculture. But at the same time, we must also actively face the difficulties encountered in the development of agricultural robots at this stage, only by crossing multiple scientific fields and promoting the development of each other, in order to truly improve machine intelligence.

2. Product market demand analysis

2.1 Product Market Demand Description

China is vigorously developing its agricultural economy, with retail sales expected to exceed 800 billion yuan by 2025. The State Council has issued multiple policy documents related to smart agriculture, aiming to promote and accelerate technological development and achieve precise breakthroughs in agricultural technology. So far, many large agricultural enterprises

in China have been actively developing and applying technology in their operations. The primary target audience for agricultural robots includes agricultural production companies, individual farmers, and agricultural research institutions. For agricultural production companies, agricultural robots can help achieve automation and intelligence in field operations, improve crop yield and quality, and reduce labor costs and resource wastage. Individual farmers can utilize agricultural robots to enhance the efficiency of field operations, reduce physical labor burdens, and increase income and quality of life. Agricultural research institutions can benefit from agricultural robots by providing an experimental platform and data support, promoting agricultural technological innovation and research progress.

2.2 Competitor analysis

In the agricultural robot market, several competitors have emerged. For example, John Deere has launched a series of agricultural robot products, including autonomous tractors and intelligent field operation solutions. Blue River Technology has developed agricultural robots based on machine vision and machine learning, capable of precise weeding and spraying operations.

2.3 Driving Factors

Efficient agricultural production: Agricultural robots can provide efficient, precise, and automated solutions for agricultural production, reducing labor burden and improving the efficiency of crop planting, harvesting, and processing.

Labor shortage: In some regions, rural labor outflow and changes in age structure have led to a shortage of agricultural labor. Agricultural robots can fill this labor gap by providing reliable automation solutions, ensuring that crop planting and harvesting are not affected by labor shortages.

Reduced production costs: Agricultural robots can reduce labor costs in the agricultural production process by optimizing resource utilization through automation, precision, and efficiency, thereby lowering production costs.

Precision agriculture management: Agricultural robots are equipped with various sensors and advanced technologies that can collect and analyze a large amount of field data, providing information on crop growth status, soil quality, and weather conditions, among others. This data can help farmers implement precision agriculture management, optimizing crop growth environments, fertilization, irrigation, and other operations, thereby increasing crop yield and quality while reducing crop loss rates.

Environmentally friendly agriculture: Agricultural robots can reduce the use of chemical pesticides and fertilizers by implementing precise targeted fertilization and pest monitoring, promoting environmentally friendly agricultural production methods.

2.4 Product Functional Requirements

The demand for agricultural robots in the market primarily stems from the need for efficient, automated, cost-effective, and environmentally friendly solutions for agricultural production. With the continuous development of agricultural technology and the growing demand for agricultural production, the agricultural robot market has vast prospects.

The product functional requirements of agricultural robots may vary depending on their application scenarios and intended tasks. Here are some common functional requirements for agricultural robot products:

Automation: Agricultural robots should possess functions such as autonomous navigation, path planning, and obstacle avoidance, enabling them to perform operations automatically in the fields, such as sowing, fertilizing, watering, weeding, and harvesting.

Sensor technology: Agricultural robots should be equipped with various sensor technologies, such as cameras, infrared, LIDAR, and ultrasonic sensors, to perceive real-time information about the agricultural environment, crop growth status, and soil quality.

Data analysis and decision support: Agricultural robots should be capable of collecting, analyzing, and storing a large amount of field data, including crop growth data, weather information, and soil analysis results. Additionally, robots should provide decision support through data analysis to assist farmers in precision agriculture management.

Precision operation and fertilization: Agricultural robots should possess precise operational capabilities to perform accurate tasks such as seeding, fertilizing,

and spraying based on crop requirements, avoiding resource waste and environmental pollution.

Remote monitoring and control: Agricultural robots should be able to connect to the internet, enabling remote monitoring and control. Farmers can manage robot operation plans, monitor crop conditions, and adjust operation parameters remotely using a mobile phone or computer.

Multi-functionality and modular design: The design of agricultural robots should feature multi-functionality and modularity, allowing flexible configuration and expansion based on different operational requirements and field conditions.

Energy management and automatic charging: Agricultural robots should have an efficient energy management system, utilizing renewable energy sources such as solar power or batteries to ensure prolonged operational capability and sustainable energy supply.

Reliability and durability: Agricultural robots should possess high reliability and durability, capable of adapting to complex field environments and harsh weather conditions.

2.5 The result of the product market demand analysis

In summary, the product functional requirements of agricultural robots primarily encompass automation, sensor technology, data analysis and decision support, precision operation, remote monitoring and control, multi-functionality design, energy management, as well as reliability and durability. These functionalities can assist farmers in enhancing agricultural production efficiency, cost savings, and achieving precision agriculture management.

3. Agricultural Harvesting Robot – Weed Recognition

3.1 Project Basis and Key Scientific Questions:

3.1.1 Project Basis

1. Efficiency and Cost Reduction: Automated harvesting robots can significantly improve agricultural productivity while reducing dependence on manual labor, especially in areas with high labor costs or labor shortages.

2.Precision Agriculture Development: With the advancement of precision agriculture, using high-tech methods to improve the yield and quality of crops has become a trend. Weed recognition technology can help robots more accurately locate and harvest tomatoes, minimizing damage to the crops.

3.Environmental Friendliness and Sustainable Development: Reducing the use of chemical herbicides and controlling weeds mechanically is beneficial for environmental protection and the development of sustainable agriculture.

3.1.2 Key Scientific Questions

1. How to achieve efficient image recognition technology to distinguish between tomatoes and weeds, especially in complex natural environments?

2. How to ensure the stability and accuracy of the recognition system under different lighting and weather conditions?

3. How to integrate the recognition system with the robot's motion control system?

3.2 Research Objectives、Content and Technical Solution

3.2.1 Research Objects

Develop an efficient weed recognition algorithm – to create an image recognition algorithm that can accurately differentiate between tomatoes and weeds, while minimizing damage to unripe tomatoes and surrounding crops, and to integrate this with the robot's movements.

3.2.2 Research Content

3.2.2.1 Data Collection and Preprocessing:

Image Data Collection: Collect a large number of images of tomatoes and weeds in field environments, covering different growth stages, lighting conditions, and backgrounds.

Data Annotation and Classification: Annotate the collected image data in detail, including identifying markers for different types of weeds and tomatoes.

Image Preprocessing: Process images, such as resizing, cropping, enhancing contrast, etc., to improve the efficiency and accuracy of subsequent processing.

3.2.2.2 Image Recognition and Machine Learning Model Development:

Feature Extraction: Extract key features from images, such as color, shape, texture, etc., which are crucial for differentiating between tomatoes and weeds.

Algorithm Selection and Training: Choose CNN machine learning algorithms to train

Model Validation and Optimization: Validate the accuracy of the model using a test set and adjust and optimize the model based on the test results.

3.2.2.3 Environmental Adaptability Research:

Performance Assessment in Different Environments: Test and evaluate the performance of the weed recognition system under various environmental conditions such as different light, humidity, soil types, etc.

Adaptability Adjustment and Optimization: Adjust the algorithm based on environmental test results to improve stability and accuracy in diverse environments.

3.2.2.4 Real-Time Data Processing and Decision Making:

Development of Fast Processing Algorithms: Develop algorithms capable of quickly processing real-time images collected in the field.

Design of Decision Logic: Construct decision logic so that the robot can make reasonable harvesting decisions based on recognition results.

3.2.2.5 System Integration and Testing:

Integration with Robot Hardware: Integrate the weed recognition system with other components of the robot (such as the mechanical arm, mobility system, etc.).

Field Trials: Conduct system tests in actual field conditions to assess the overall system performance and reliability.

3.2.3 Technical Solution

The technical solution for weed recognition consists of the following components:

3.2.3.1 Image Acquisition System:

High-Resolution Cameras: Use high-resolution cameras to capture images in the field. These cameras should work under different lighting conditions and have enough dynamic range to capture details.

3.2.3.2 Multi-Angle Shooting:

Design the robot to take pictures of plants from multiple angles to ensure comprehensive capture of tomatoes and weeds.

3.2.3.3 Data Processing and Feature Extraction:

Preprocessing: Preprocess the collected images for noise reduction, contrast enhancement, etc., to improve the efficiency and accuracy of subsequent processing.

Feature Extraction: Use image processing techniques to extract key features, such as color, shape, texture, etc., to distinguish between tomatoes and weeds.

3.2.3.4 Machine Learning and Image Recognition:

- Machine Learning Model: Convolutional Neural Network (CNN).

- Training and Optimization: Train the model using a dataset of labeled tomato and weed images, and adjust and optimize based on the results of validation.

3.2.3.5 Environmental Adaptability Optimization:

- Environmental Testing: Test the performance of the recognition system under different environmental conditions (such as different weather, lighting, and soil types).

- Algorithm Adjustment: Adjust algorithm parameters based on test results to ensure stability and accuracy in varying environments.

3.2.3.6 Real-Time Decision-Making and Control Systems:

- Real-Time Data Processing: Develop a system capable of quickly processing collected images and making recognition decisions.

3.3 Innovation

1. Selection of object detection algorithm: YOLOR has its unique advantages compared with other real-time object detection algorithms (e.g., Faster R-CNN, SSD, etc.), such as fast speed, high accuracy, simple structure. These features make YOLO a competitive object detection algorithm, and it has achieved good results in many real-world application scenarios.

2.Data preprocessing techniques: data is key for any machine learning task. When performing obstacle recognition, a variety of problems may be encountered, such as changing lighting conditions, changing viewing angles, and occlusion. By reasonably preprocessing the image data, these problems can be improved, thus increasing the accuracy of the detection results. YOLOv8 supports a variety of data preprocessing methods, so you can choose the most suitable method for data preprocessing according to the actual situation.

3. Selection of training parameters: YOLOv8 has many adjustable training parameters, such as learning rate, batch size, optimizer type, etc. By carefully selecting these parameters, you can further improve the accuracy of the detection results. By carefully selecting these parameters, the performance of YOLOv8 can be further improved. This needs to be selected and debugged according to the specific dataset and the purpose of the experiment.

4. Diversity of Obstacle Types: There are many types of obstacles, both static ones such as walls, furniture, etc., and dynamic ones such as pedestrians and vehicles. For different types of obstacles, different detection strategies may be needed, for example, for smaller obstacles, the detection threshold can be increased to reduce false detections; for larger obstacles, the detection accuracy can be improved by using a larger sensing field.

5. Multi-modal fusion: In addition to visual sensors, other types of sensors can be used to obtain environmental information, such as ultrasonic sensors, LIDAR, etc. By fusing these multi-modal information, a more accurate and comprehensive environmental sensing capability can be obtained, thus improving the performance of obstacle recognition.

6. Application of Heterogeneous Computing Platforms: With the development of hardware, there are now a variety of heterogeneous computing platforms to choose from, such as GPUs, FPGAs, ASICs, etc., which all have their own advantages and disadvantages. By reasonably selecting and configuring heterogeneous computing platforms, the performance of computer vision system can be significantly improved.

7. Real-time requirements: for mobile robots, real-time is very important. This requires that the system be designed with consideration of how quickly the task can be accomplished and how to effectively utilize the limited computational resources. This can be achieved by optimizing the code to achieve this, or by adopting some real-time operating system support.

3.4 Project Overview

3.4.1 Background

Obstacle recognition has important applications in fields such as autonomous driving, intelligent robotics and UAVs. This project aims to realize an efficient and accurate obstacle detection and recognition system based on the YOLO v8 (You Only Look Once version 8) model.

3.4.2 Objectives

1) Develop a model that can accurately recognize and locate obstacles in complex scenes.

2) Improve the real-time and efficiency of the model to meet the requirements of practical applications.

3) Conduct robustness tests under different environmental conditions to ensure that the model can work stably in various situations.

3.4.3 Methods and techniques

3.4.3.1 YOLO v8 model selection

The YOLO v8 model is chosen as the base model due to its excellent performance and real-time performance in the target detection task.

3.4.3.2 Data set preparation

Collect and organize obstacle image datasets containing various scenes, weather and lighting conditions. Ensure the diversity of the dataset to improve the generalization ability of the model.

3.4.3.3 Data Preprocessing

Preprocess the data, including image enhancement, resizing and labeling to improve model training.

3.4.3.4 Model Training

The YOLO v8 model is trained using the prepared dataset. Adjust the hyperparameters to optimize the performance of the model. Utilize pre-trained models to speed up the training process.

3.4.3.5 Model Optimization

Model optimization for real-time and accuracy. Deep model compression, quantization and other techniques

are used to ensure that the model can run efficiently even on embedded devices.

3.4.4. Application and Deployment

3.4.4.1 Real Scene Testing

Test in real scenarios to verify the stability and robustness of the model in real environments.

3.4.4.2 Deployment Program

Design the deployment scheme of the model and consider how the model can be deployed on edge devices or embedded systems to realize the actual application.

3.5 Achievement Presentation

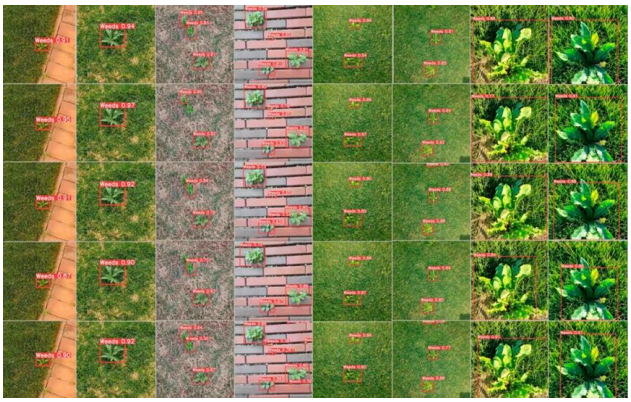

Figure.1

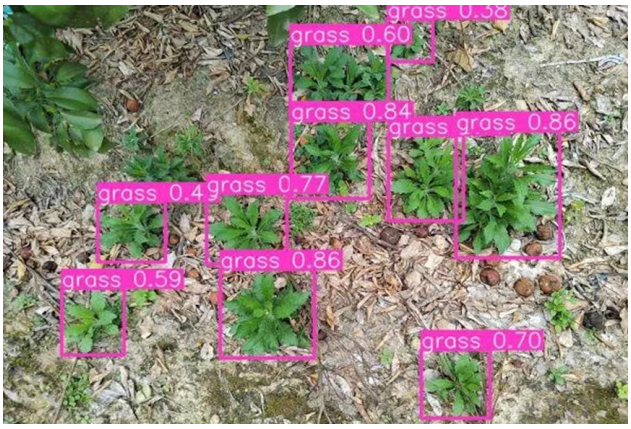

Figure.2

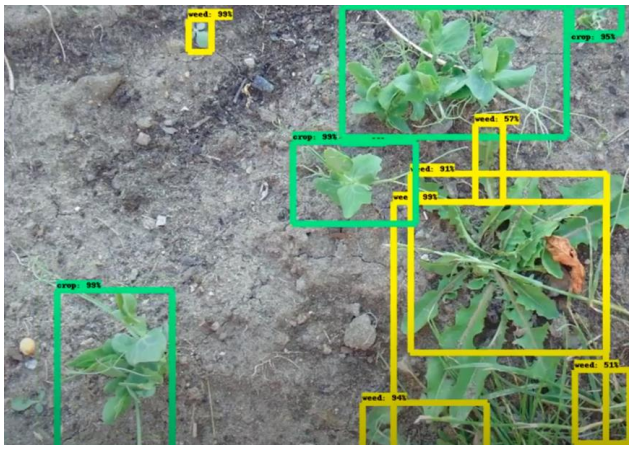

Figure.3

4. Conclusion

Based on the market and technical analysis conducted, the global market potential for agricultural robots and precision agriculture solutions is significant. Despite the large market demand, accurate and reliable weed recognition remains a persisting challenge hampering the wide adoption of automated harvesting robots.

5. Personal gains

Example 1: From Xinyao. Liu

In this project, I was responsible for the development of human-computer interaction software, and in the early and middle stages of the project, I learned the subject knowledge related to robots. Finally, through learning and using Qtcreator, the development of interactive software was preliminarily completed. However, due to limited time, the software has not been further optimized and aesthetically designed. In July and August, during the summer school of the project, I, who had never been in contact with the subject of robotics, preliminarily learned the practical operation of robots and related programming under the leadership of my supervisor and graduate students. In the process, my programming skills also improved. During the later completion of the project, I learned the use of qtcreator software and knowledge about UI design by myself. However, if I want to develop a software that interacts with robots, I need to use all kinds of software related to robots and apply it in software development, which is the challenge I face when developing software. Finally, with the concerted efforts of the project team members, we completed the preliminary development of the human-computer interaction software. I believe that with the

improvement of our professional ability, we can make more complete human-computer interaction software in the future.